Project 1 Report

Liam Stevens, Cassini Rajesh, Adam Skutnik

AERSP 304

3/7/2025

Contributions

Liam Stevens: 33%

Cassini Rajesh: 33%

Adam Skutnik: 33%

Honor Statement

I, Liam Stevens, have neither given nor received any assistance on this project.

I, Cassini Rajesh, have neither given nor received any assistance on this project.

I, Adam Skutnik, have neither given nor received any assistance on this project.

Introduction

The circular three-body problem is one where the goal is to determine the trajectory of three separate point masses that orbit each other. Although it is seemingly similar to the two-body problem, the third mass adds another level of complexity. However, the three-body problem is very vital to simulate realistic scenarios. In this case, a spacecraft and two primary bodies such as planets or stars. This scenario is depicted below in Figure 1.

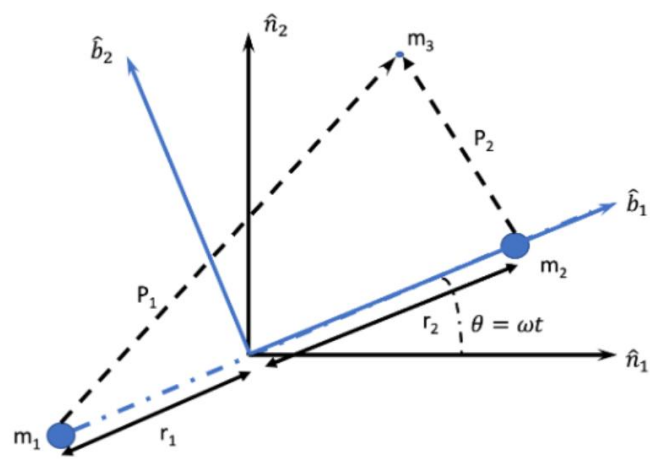


Figure 1. *Schematic of a spacecraft and two larger bodies.*

Figure 1 depicts a three-body problem in which m_1 and m_2 are the larger bodies such as the Earth and Moon, while m_3 is the smaller body such as a spacecraft. Here the N frame is the inertial frame while B is the rotating body frame in which masses 1 and 2 stay in line with b_1. The two larger masses are separated from the origin by a distance r which is comprised of r_1 and r_2.

Equations of Motion

To accurately represent the motion of the spacecraft m_3 it is required to determine the kinetic and potential energy so that a Lagrangian approach may be taken. The potential energy is given in equation 1 below.

$$V = -G\frac{m_3 m_1}{P_1} - G\frac{m_3 m_2}{P_2} \tag{1}$$

The kinetic energy can be found by taking the derivative, equation 3, of the spacecraft's position vector, equation 2, as shown below.

$$r_3 = xb_1 + yb_2 \tag{2}$$

$$v_3 = (x' - y\omega)b_1 + (y' - x\omega)b_2 \tag{3}$$

In order to fully determine the Lagrangian, P_1 and P_2 need to rewritten into terms of x and y which is provided below in equations 4 and 5.

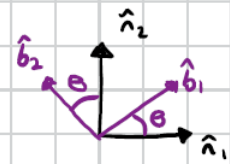
$$P_1 = \sqrt{(x + \mu)^2 + y^2} \tag{4}$$

$$P_2 = \sqrt{(1 - x - \mu)^2 + y^2} \tag{5}$$

The full work done to determine the equations of motion is provided below.

1)

a) Mass 3 $\vec{r} = x\hat{b}_1 + y\hat{b}_2$



$\theta = \omega t$

$\dot{\theta} = \omega$

$\hat{b}_1 = c\theta\,\hat{n}_1 + s\theta\,\hat{n}_2$

$\hat{b}_2 = -s\theta\,\hat{n}_1 + c\theta\,\hat{n}_2$

$\hat{b}_3 = \hat{b}_3$

$$\dot{\vec{r}} = \dot{x}\hat{b}_1 + x\dot{\hat{b}}_1 + \dot{y}\hat{b}_2 + y\dot{\hat{b}}_2$$

$$\dot{\hat{b}}_1 = \omega_{B/N} \times \hat{b}_1$$

$$\dot{\hat{b}}_1 = \dot{\theta}\hat{b}_3 \times \hat{b}_1 = \dot{\theta}\hat{b}_2$$
$$= \omega\hat{b}_2$$

$$\dot{\hat{b}}_2 = \omega_{B/N} \times \hat{b}_2$$
$$= \dot{\theta}\hat{b}_3 \times \hat{b}_2$$
$$= -\omega\hat{b}_1$$

$$\dot{\vec{r}} = \dot{x}\hat{b}_1 + x(\omega\hat{b}_2) + \dot{y}\hat{b}_2 + y(-\omega\hat{b}_1)$$

$$\dot{\vec{r}} = \dot{x}\hat{b}_1 - y\omega\hat{b}_1 + x\omega\hat{b}_2 + \dot{y}\hat{b}_2$$

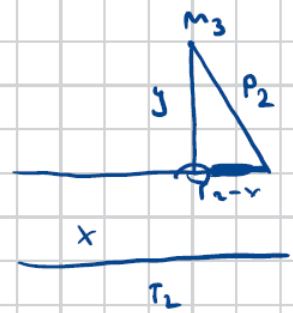
$$\dot{\vec{r}} = (\dot{x} - y\omega)\hat{b}_1 + (x\omega + \dot{y})\hat{b}_2$$

$$T = \frac{1}{2}|\dot{\vec{r}}|^2 = \frac{1}{2}\left[(\dot{x} - y\omega)^2 + (x\omega + \dot{y})^2\right], \quad V = -G\frac{m_3 m_1}{\rho_1} - G\frac{m_3 m_2}{\rho_2}$$

Using the position vector of m_3 in the b frame, the velocity vector with respect to time is found. This velocity vector directly corresponds to the kinetic energy T.

$\mathcal{L} = T - V$

Convert P to x and y



$$y^2 + (r_2 - x)^2 = P_2$$

$$P_2^2 = y^2 + r_2^2 - 2xr_2 + x^2$$

$$P_2 = \sqrt{(r_2 - x)^2 + y^2}$$

$$\vec{P}_1 = (r_1 + x)\hat{b}_1 + y\hat{b}_2$$

$$P_1 = \sqrt{(r_1 + x)^2 + y^2}$$

$$(\dot{x} - y\omega)^2 = \dot{x}^2 - 2\omega\dot{x}y + y^2\omega^2$$

$$(x\omega + \dot{y})^2 = x^2\omega^2 + 2\omega x\dot{y} + \dot{y}^2$$

After obtaining the kinetic energy equation, P_1 and P_2 are solved in terms of x and y coordinates in the b frame. This is done to expand the potential energy V given in equation 1 and, in turn, fully determine the Lagrangian.

$$\mathcal{L} = \frac{1}{2} m_3 \left[(\dot{x} - y\omega)^2 + (x\omega + \dot{y})^2 \right] + \frac{G m_3 m_1}{\sqrt{(r_1 + x)^2 + y^2}} + \frac{G m_3 m_2}{\sqrt{(r_2 - x)^2 + y^2}}$$

$$\mathcal{L} = \frac{1}{2} m_3 \left[(\dot{x}^2 - 2\omega \dot{x} y + y^2 \omega^2) + (x^2 \omega^2 + 2\omega x \dot{y} + \dot{y}^2) \right] + \frac{G m_3 m_1}{\sqrt{(\mu + x)^2 + y^2}} + \frac{G m_3 m_2}{\sqrt{(1 - \mu - x)^2 + y^2}}$$

For x:

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{x}}\right) - \frac{\partial \mathcal{L}}{\partial x} = 0$$

$$\frac{\partial \mathcal{L}}{\partial \dot{x}} = \dot{x} - 2\omega y \rightarrow \frac{d}{dt}(\dot{x} - \omega y) = \ddot{x} - \omega \dot{y}$$

$$\frac{\partial \mathcal{L}}{\partial x} = x\omega^2 + \omega \dot{y} - \frac{G m_3 m_1 (\mu + x)}{((\mu + x)^2 + y^2)^{3/2}} + \frac{G m_3 m_2 (-x + 1 - \mu)}{((1 - \mu - x)^2 + y^2)^{3/2}}$$

$$m_3 (\ddot{x} - 2\omega \dot{y} - x\omega^2) = \frac{-G m_3 m_1 (\mu + x)}{((\mu + x)^2 + y^2)^{3/2}} + \frac{G m_3 m_2 (-x + 1 - \mu)}{((1 - \mu - x)^2 + y^2)^{3/2}}$$

$$\ddot{x} - 2\omega \dot{y} - x\omega^2 = \frac{-G m_1 (\mu + x)}{\rho_1^3} - \frac{G m_2 (x - 1 + \mu)}{\rho_2^3}$$

Using that Lagrange equation, in which L equals kinetic energy minus potential energy, the full equation needed to solve for the x and y equations of motion is obtained.

$$\mu (m_1 + m_2) = m_2 \rightarrow (m_1 + m_2) = \frac{m_2}{\mu}$$

$$\frac{\omega^2 (r_1 + r_2)^3}{G} = m_1 + m_2$$

$$\frac{\omega^2 \mu}{G} = m_2$$

$$\omega^2 = \frac{G (m_1 + m_2)}{(r_1 + r_2)^3}$$

$$\mu = \frac{m_2}{m_1 + m_2}$$

$$\omega^2 = G (m_1 + m_2)$$

$$m_1 = \frac{\omega^2}{G} - m_2 = \frac{\omega^2}{G} - \frac{\omega^2 \mu}{G} = \frac{\omega^2 - \omega^2 \mu}{G} = \frac{\omega^2 (1 - \mu)}{G}$$

$$\ddot{x} - 2\omega \dot{y} - x\omega^2 = -\omega^2 \frac{(1 - \mu)(x + \mu)}{\rho_1^3} - \omega^2 \frac{\mu (x - 1 + \mu)}{\rho_2^3} \quad (2)$$

After obtaining the expression for the time derivative of the partial derivative of x' (the same as x dot) minus the partial derivative of x, the formula for omega squared and mu to simplify to the final form of equation two is found.

For y:

$$\mathcal{L} = \frac{1}{2} m_3 \left[(\dot{x} - y\omega)^2 + (x\omega + \dot{y})^2 \right] + \frac{G m_3 m_1}{\sqrt{(r_1 + x)^2 + y^2}} + \frac{G m_3 m_2}{\sqrt{(r_2 - x)^2 + y^2}}$$

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{y}}\right) - \frac{\partial \mathcal{L}}{\partial y} = 0$$

$$\frac{\partial \mathcal{L}}{\partial \dot{y}} = m_3 (x\omega + \dot{y})$$

$$\frac{d}{dt}\left(m_3 x\omega + m_3 \dot{y}\right) = m_3 \dot{x}\omega + m_3 \ddot{y}$$

$$\frac{\partial \mathcal{L}}{\partial y} = m_3(-\omega\dot{x} + \omega^2 y) - \frac{y G m_3 m_1}{((r_1 + x)^2 + y^2)^{3/2}} - \frac{y G m_3 m_2}{((r_2 - x)^2 + y^2)^{3/2}}$$

$$\ddot{y} + 2\omega\dot{x} - \omega^2 y = -\frac{y G m_1}{\rho_1^3} - \frac{y G m_2}{\rho_2}$$

$$\ddot{y} + 2\omega\dot{x} - \omega^2 y = -\omega^2 \frac{(1-\mu) y}{\rho_1^3} - \omega^2 \frac{\mu y}{\rho_2^3} \quad (3)$$

Once again, the Lagrange equation is used, but this time for the y equation of motion. The equation is simplified using the omega squared and mu equations to finally arrive at the final result for equation three.

$$\frac{d}{dt} x(t) = \frac{d}{d\tau} x\left(\frac{1}{\omega}\tau\right) = x' \frac{d\tau}{dt}$$

$$\dot{x} = x' \cdot \frac{d\tau}{dt} = x' \cdot \omega$$

$$\frac{d^2}{dt^2} x(t) = \frac{d^2}{d\tau^2} x\left(\frac{1}{\omega}\tau\right) = \frac{d}{d\tau}\left(\frac{d}{d\tau} x\left(\frac{1}{\omega}\tau\right)\right) = \frac{d}{d\tau}\left(\frac{d}{d\tau} x(\tau) \cdot \omega\right) = \frac{d^2}{d\tau} x(\tau)\omega^2$$

For $v \equiv \frac{d}{\tau} x(\tau)$

$$\frac{d}{d\tau} v = \frac{d}{d\tau} v \frac{d\tau}{dt} = \frac{d}{d\tau} v \cdot \omega$$

$$x'' - 2y' - x = -\frac{(1-\mu)(x+\mu)}{\rho_1^3} - \frac{\mu(x - 1 + \mu)}{\rho_2^3} \quad (4)$$

Using the scaled time variable, relationships for x double dot, y dot, and x are found in order to get rid of omega, arriving at equation four.

$$\frac{d}{dt}y(t) = \frac{d}{d\tau}y\left(\frac{1}{\omega}\tau\right) = y' \cdot \frac{d\tau}{dt}$$

$$\dot{y} = y' \cdot \frac{d\tau}{dt} = y' \cdot \omega$$

$$\frac{d^2}{dt^2}y(t) = \frac{d^2}{d\tau^2}y\left(\frac{1}{\omega}\tau\right) = \frac{d}{d\tau}\left(\frac{d}{d\tau}y\left(\frac{1}{\omega}\tau\right)\right) = \frac{d}{d\tau}\left(\frac{d}{d\tau}y(\tau)\cdot\omega\right) = \frac{d^2}{d\tau}y(\tau)\omega^2$$

for $v = \frac{d}{d\tau}y(\tau)$

$$\frac{d}{d\tau}v = \frac{d}{d\tau}v\,\frac{d\tau}{dt} = \frac{d}{d\tau}v \cdot \omega$$

$$y'' + 2x' - y = -\frac{(1-\mu)y}{\rho_1^3} - \frac{\mu y}{\rho_2^3} \quad (5)$$

$$U = \frac{1}{2}(x^2+y^2) + \frac{1-\mu}{\rho_1} + \frac{\mu}{\rho_2}$$

$$\rho_2 = \sqrt{(r_2-x)^2+y^2}$$

$$\rho_1 = \sqrt{(r_1+x)^2+y^2}$$

Following the same path and using the scaled time variable, the relationships for y double dot, x dot, and y are derived to get rid of omega, thus arriving at equation five.

$$\frac{\partial u}{\partial x} = x - \frac{1-\mu}{\rho_1^3}(x+\mu) - \frac{\mu}{\rho_2^3}(x-1+\mu)$$

$$\frac{\partial u}{\partial y} = y - \frac{1-\mu}{\rho_1^3}y - \frac{\mu}{\rho_2^3}y$$

$$x'' - 2y' = x - \frac{1-\mu(x+\mu)}{\rho_1^3} + \frac{\mu(x-1+\mu)}{\rho_2^3}$$

$$x'' - 2y' = \frac{\partial u}{\partial x} \quad (7)$$

$$y'' + 2x' = y - \frac{(1-\mu)y}{\rho_1^3} - \frac{\mu y}{\rho_2^3}$$

$$y'' + 2x' = \frac{\partial u}{\partial y} \quad (8)$$

After obtaining equations four and five, these equations are further simplified to arrive at equations seven and eight by taking the partial derivative of the effective potential with respect to x and y. Thus, the equations of motion for the system are given below.

$$x'' - 2y' - x = -\frac{(1-\mu)(x+\mu)}{P_1^3} - \frac{\mu(x-1+\mu)}{P_2^3} \tag{6}$$

$$y'' + 2x' - y = -\frac{(1-\mu)y}{P_1^3} - \frac{\mu y}{P_2^3} \tag{7}$$

In these equations of motion, the μ represents the mass ratio of the system. This ratio is further elaborated upon below in equation 8.

$$\mu = \frac{m_2}{m_1+m_2} \tag{8}$$

Equilibrium Points

From the discovered equations of motions from the previous section, the equilibrium points are determinable. These points are assumed to be at rest in the rotating frame. In other words, all derivatives of x and y are equal to zero. In the three-body problem, there 5 equilibrium points, or Lagrange points, that are discovered. These points are depicted below in Figure 2.

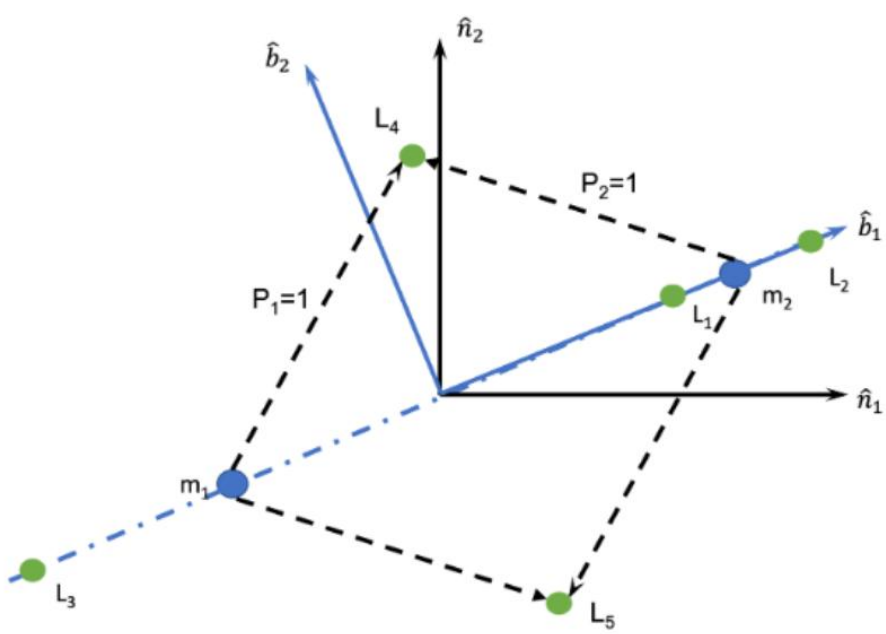


Figure 2. *A schematic that depicts where the 5 equilibrium points reside in the system.*

As seen in the figure, 3 of these equilibrium points lie along the b_1 vector (where y=0). The two other points, L_4 and L_5 form equilateral triangles with the two primary bodies. Therefore, they can be found by equating P_1 to P_2 and plugging in the respected mass ratio μ as shown below in equation 8.

$$\sqrt{(x+\mu)^2+y^2}=\sqrt{(1-x-\mu)^2+y^2} \tag{8}$$

Since there are two unknowns and two equations, the coordinates of these Lagrangian points can be found. The values for L_4 and L_5 are given below in Table 1.

Table 1. Values of L_4 and L_5

System	**L_4 (x, y)**	**L_5 (x, y)**
Sun-Earth	0.499997, 0.866025	0.499997, -0.866025
Earth-Moon	0.487849, 0.866025	0.487849, -0.866025
Saturn-Titan	0.499645, 0.866094	0.499645, -0.866094

As seen in the table, the x-values of L_4 and L_5 are equal. However, their y-values are of different signs although equal in magnitude. This is the expected result, as observed in Figure 2 where L_4 and L_5 lie on opposite sides of the b_1 vector.

Stability of Lagrange Points

With these found Lagrange points, it is now vital to determine their stability. This is done by linearizing the equations of motion as seen below.

c)

$$x'' - 2y' = U_x \rightarrow F(x'', y', x) = x'' - 2y' - U_x$$

$$F_1(x'', y', x) = F(x''_0, y_0', x_0) + \frac{\partial F}{\partial x''}\Big|_{x''=x_0''} \delta x'' + \frac{\partial F}{\partial y'}\Big|_{y'=y_0'} \delta y' + \frac{\partial F}{\partial x}\Big|_{x=x_0} \delta x$$

$$\frac{\partial F}{\partial x'} = 1$$

$$\frac{\partial F}{\partial y'} = -2$$

$$\frac{\partial F}{\partial x} = \frac{\partial^2 U}{\partial x^2}$$

$$(1)\ \delta x'' - 2\delta y' - \frac{\partial^2 U}{\partial x^2}\delta x - \frac{\partial^2 U}{\partial x \partial y}\delta y = 0$$

$$F_2(y'', x', y) = F(y_0'', x_0', y_0) + \frac{\partial F}{\partial y''}\Big|_{y''=y_0''} \delta y'' + \frac{\partial F}{\partial x'}\Big|_{x'=x_0'} \delta x' + \frac{\partial F}{\partial y}\Big|_{y=y_0} \delta y$$

$$\delta y'' + 2\delta x' - \frac{\partial^2 U}{\partial y^2}\delta y - \frac{\partial^2 U}{\partial x \partial y}\delta x = 0$$

$$\left.\begin{array}{l} (i)\ \delta x'' = U_{xx}\delta x + U_{xy}\delta y + 2\delta y' \\ (ii)\ \delta y'' = U_{yy}\delta y + U_{xy}\delta x - 2\delta x' \end{array}\right\} \rightarrow \begin{bmatrix} \delta x' \\ \delta y' \\ \delta x'' \\ \delta y'' \end{bmatrix} = A \begin{bmatrix} \delta x \\ \delta y \\ \delta x' \\ \delta y' \end{bmatrix}$$

$$\delta \bar{x}' = A\delta\bar{x}, \quad \delta\bar{x} = \{\delta x\ \delta y\ \delta x'\ \delta y'\}^T$$

Here, the Taylor polynomial method is used to linearize equations seven and eight. Once linearized, a system of equations is arranged to obtain the A matrix and satisfy equations 11 and 12.

To understand the stability of the Lagrange points, the equations for U_{xx}, U_{yy}, and U_{xy}, must be derived first. For this to be done, the corresponding equations for P_1 and P_2

have to be used. In this case, the P_1 and P_2 equations found earlier correspond to Lagrange points 1, 2, and 3. Once the second derivative equations of U are found, they can be put into the 4x4 A matrix that was produced. From this matrix, the eigenvalues for each point can be found by plugging in the corresponding values. Finding the eigenvalues was done through a MATLAB code due to the matrix's size and the simplicity of being able to plug in values. Finally, once these eigenvalues are found, they can be tested for stability which is provided in Table 2 below.

Table 2. Stability of Lagrange Points

	L1	**L2**	**L3**	**L4**	**L5**
Sun-Earth	Unstable	Unstable	Unstable	Stable	Stable
Earth-Moon	Unstable	Unstable	Unstable	Stable	Stable
Saturn-Titan	Unstable	Unstable	Unstable	Stable	Stable

As seen from Table 2, even though Lagrangian points 1, 2, and 3 are in equilibrium, they are unstable while Lagrangian points 4 and 5 are stable no matter the system. Points 1, 2, and 3 are unstable since their positions cause the spacecraft to be more influenced by the mass of one primary body more than the other. While they are in equilibrium, they are more susceptible to being thrown off balance.

Simulating Orbits Around L_2 In The Earth-Moon System

To simulate a spacecraft's orbit around Lagrangian point 2 in an Earth-Moon system, it's motion in x and y are plotted against each other. This motion can either be plotted in the body frame or the inertial frame. The body frame is one where the frame moves with the object, in this case the spacecraft. The inertial frame is one where the frame does not move with the object. The orbit of the spacecraft around Lagrangian point 2 in a body frame is shown below in Figure 3.

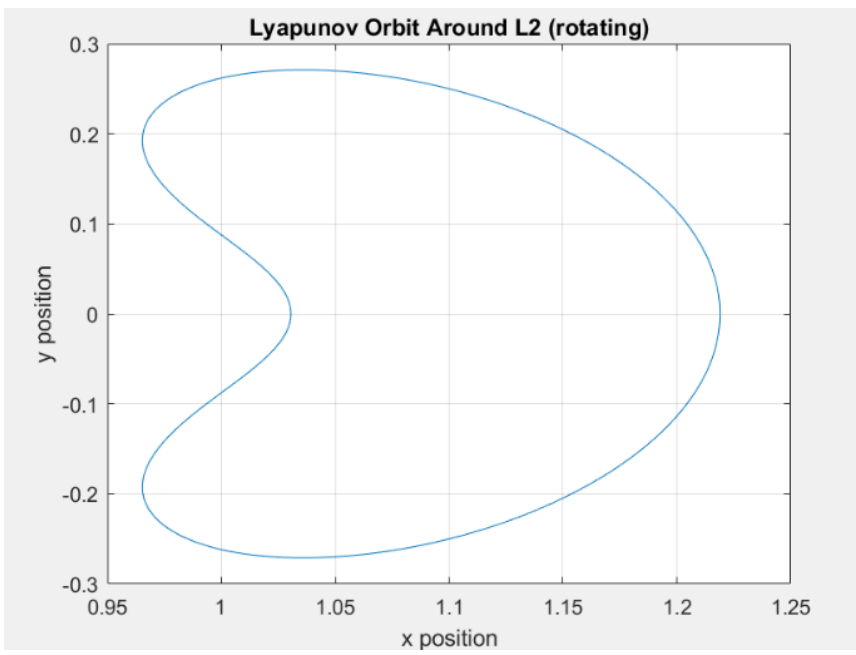


Figure 3. Graph of the Lyapunov Orbit around Lagrange point 2 in a body frame.

As seen in Figure 3, the orbit of the spacecraft is somewhat an elliptical shape where one side of the ellipse is pushed in towards the center creating a heart shaped orbit. This shape is caused by the differences of mass between the Earth and the Moon. The section of the orbit that gets pushed in corresponds to the Moon oriented part of the satellite's path in which the satellite is closer in proximity to the moon. Compare this to the inertial frame in figure 4 below.

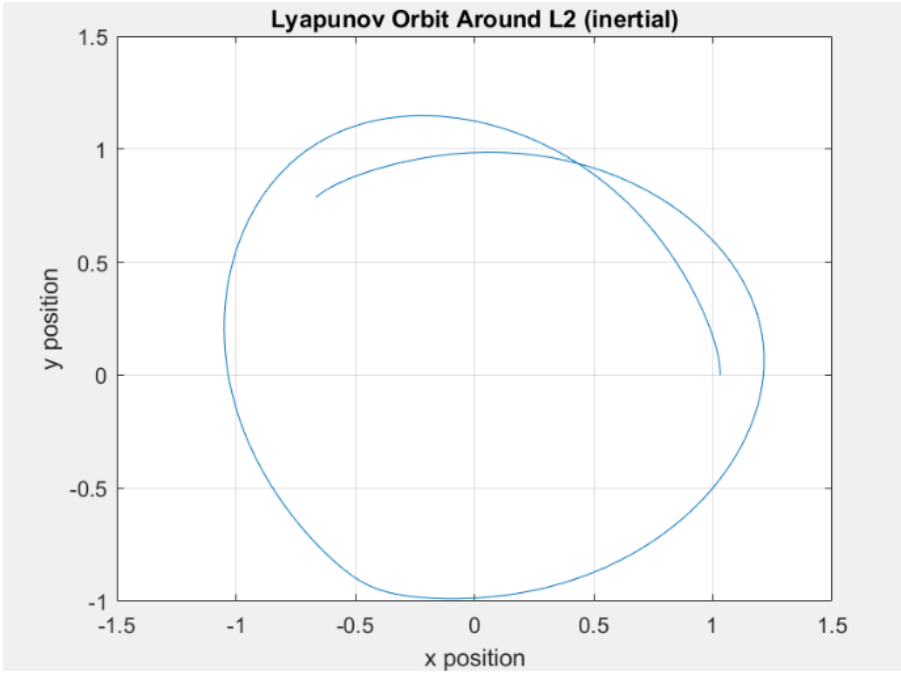


Figure 4. Graph of the Lyapunov Orbit around Lagrange point 2 in an inertial frame.

As seen in the inertial frame, the orbit nearly makes a complete circle. However, it goes off path and ends up being imperfect. With celestial bodies such as planets and stars, there are perturbations, or deviations in a body's course that are caused by gravity pull from other bodies. The three-body problem is not exempt from these perturbations. The perturbations of the Earth-Moon system are provided below in Figures 5 and 6.

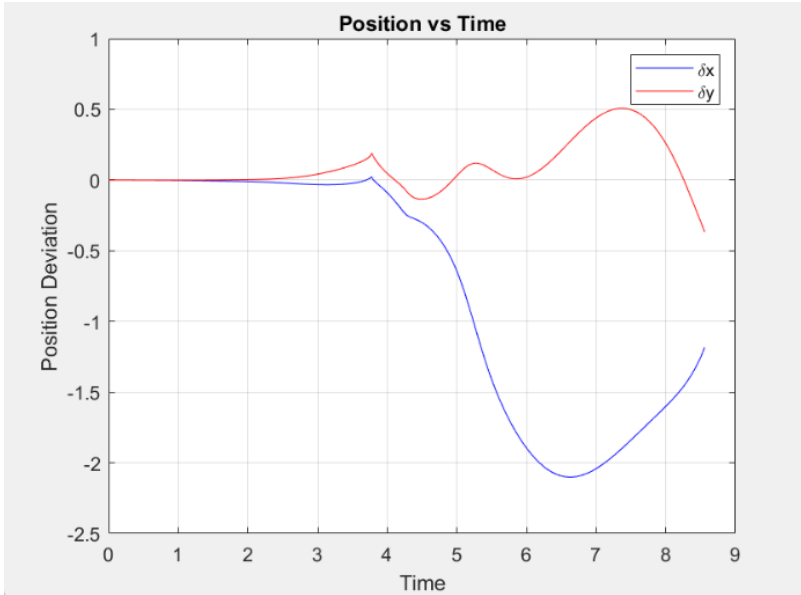


Figure 5. Perturbation in position over time for point 2.

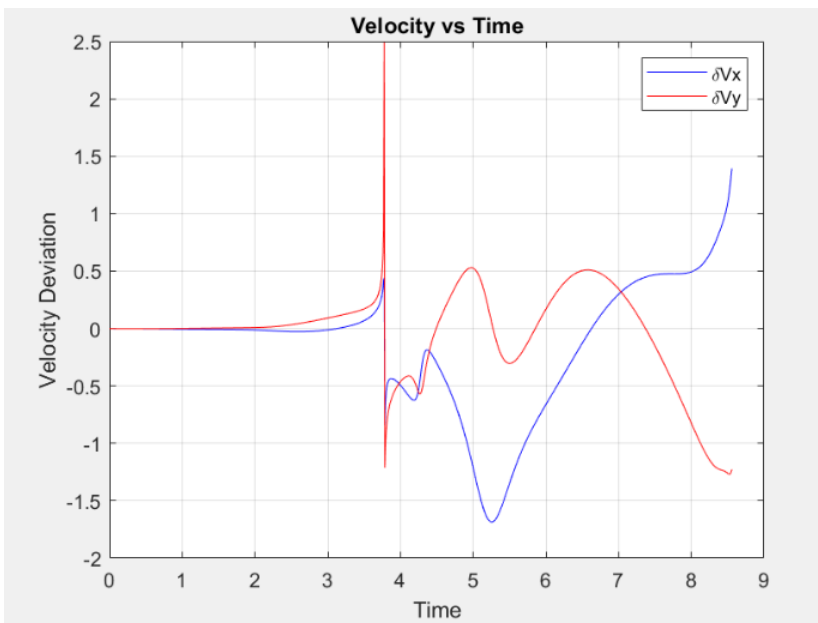


Figure 6. Perturbation in velocity over time for point 2.

As seen in the figures above, the disturbances are initially nonexistent, but over time they ramp up and cause the system to be thrown off. This tracks when re-examining the stability of Lagrange point 2. The perturbation in the neighborhood of Lagrange point 2 is given by Figure 7 below.

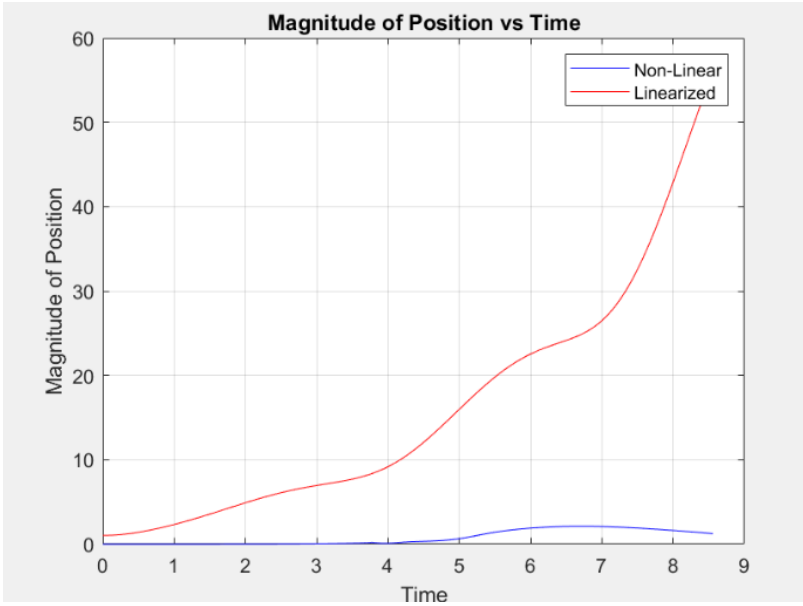


Figure 7. Perturbation in the neighborhood of point 2.

As shown in Figure 7, the change in position over time is different for the linearized and non-linearized cases, which is expected. Since the deviance of the orbit is in an oscillating manner, it is difficult to capture it in a linearized form. As the graph shows, the linearized form has disturbances from the initial starting time, and they begin to grow exponentially. On the other hand, the non-linear form stays relatively undisturbed and even when disturbances begin to show, they oscillate around the starting value.

Simulating Orbits Around L_4 in The Earth-Moon System

Both Lagrangian points 2 and 4 are being depicted in an Earth-Moon system. However, these two points result in very different orbits. The orbits around Lagrangian point 4 in both a body and an inertial frame are depicted below in Figures 8 and 9 respectively.

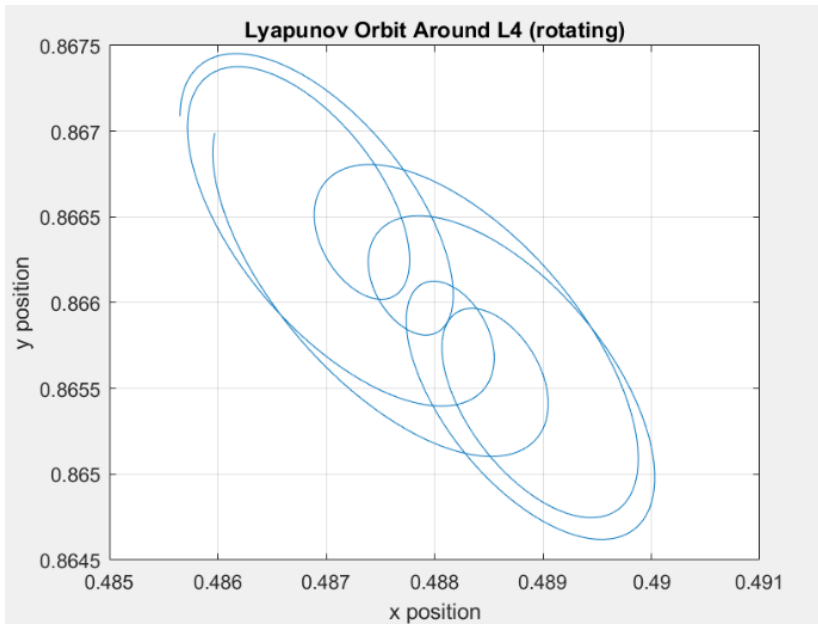


Figure 8. Graph of the Lyapunov Orbit around Lagrange point 4 in a body frame.

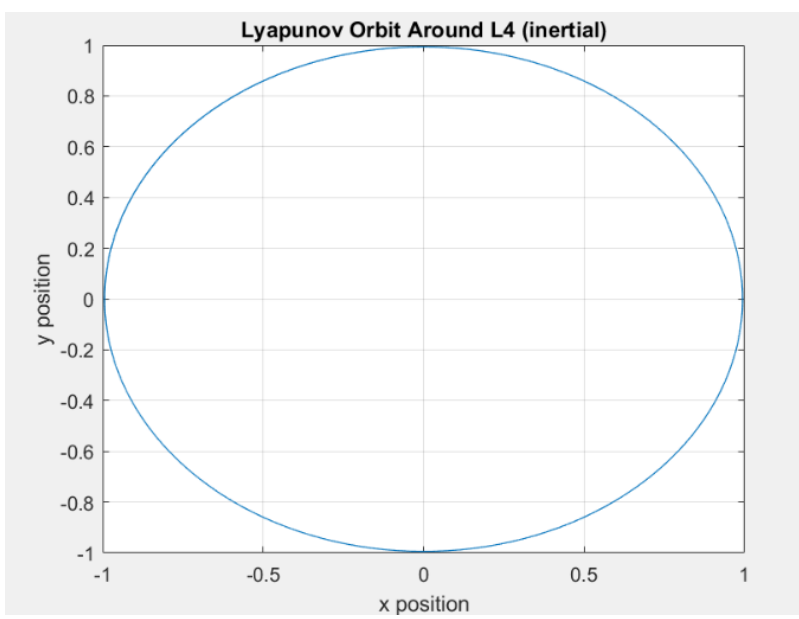


Figure 9. Graph of the Lyapunov Orbit around Lagrange point 4 in an inertial frame.

As shown in the figures above, the orbit for Lagrange point 4 in the body frame is more complex than that of point 2's orbit. Also, the inertial frame orbit for point 4 creates a complete ellipse without any inconsistency. Again, this system is also subjected to perturbations caused by the gravitational pull of other bodies. The perturbations of Lagrange point 4 are shown below in Figures 10 and 11 below.

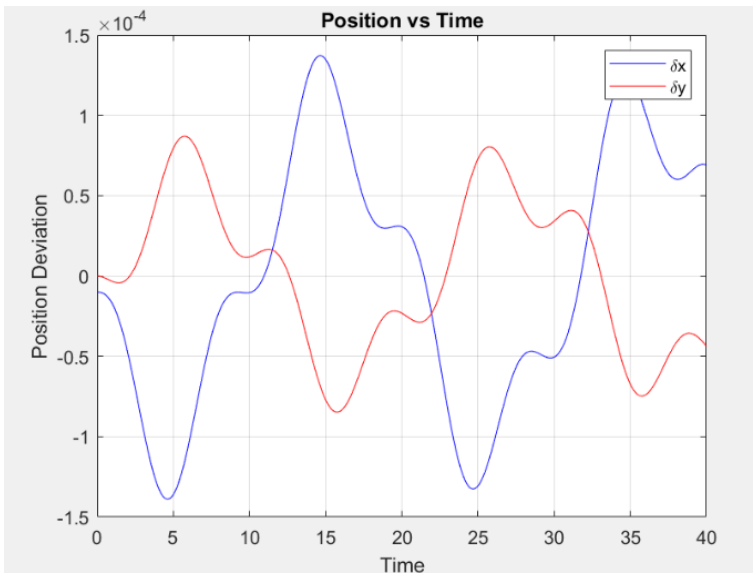


Figure 10. Perturbation in position over time for point 4.

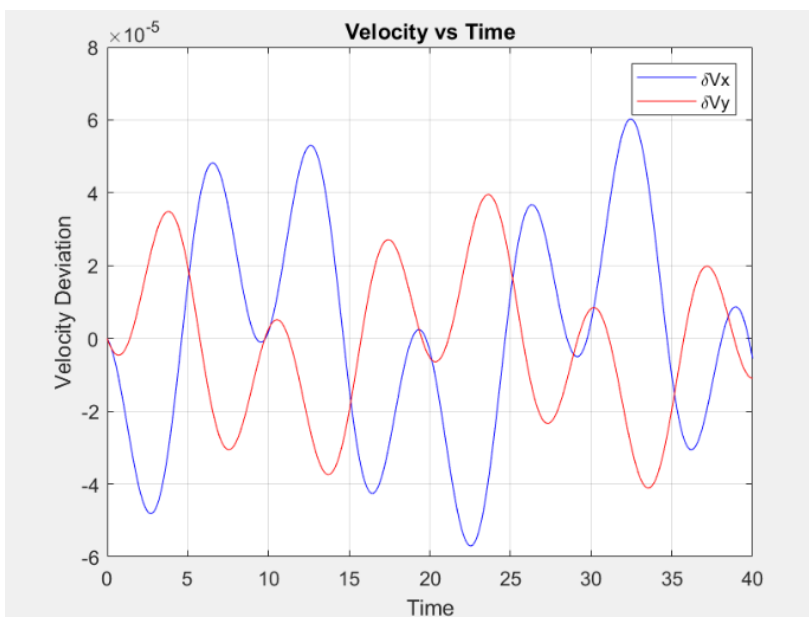


Figure 11. Perturbation in velocity over time for point 4.

Compared to the perturbations of point 2, point 4 has a more consistent and predictable pattern. Although there are still deviations, the disturbances are understandable and tend to oscillate around a set value. The perturbation in the neighborhood of Lagrange point 4 is given by Figure 12 below.

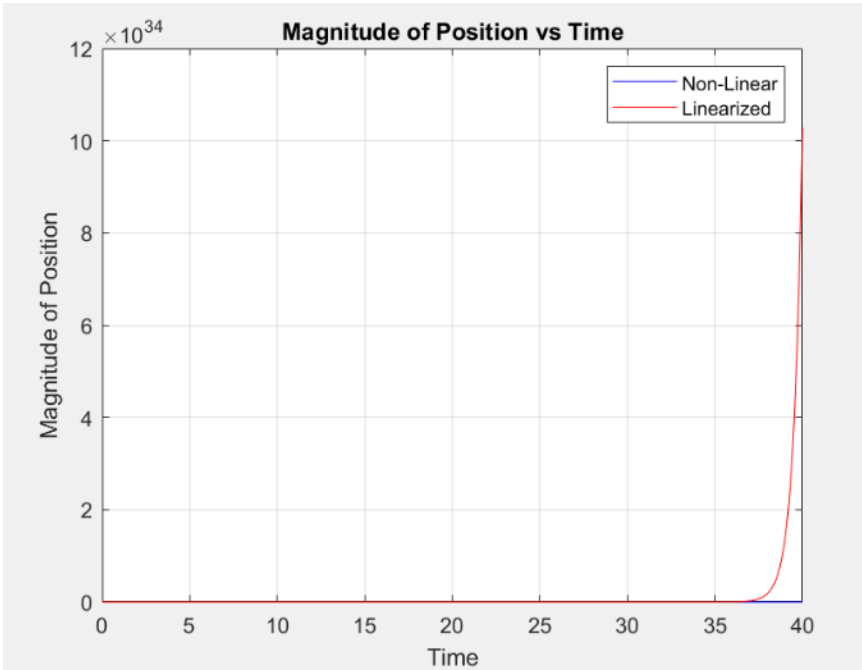


Figure 12. Perturbation in the neighborhood of point 4.

As seen in Figure 12, the linearized version remains very similar to the non-linearized version for a longer time than what was seen for point 2. This is also the expected case as Lagrange point 4 is a stable equilibrium point. This means that any perturbation will work itself out and will not grow and cause instability in the system.